MACHINE LEARNING · TECHNICAL REPORT PRESENTATION

House Price Prediction

Using Machine Learning Regression Models

An end-to-end regression pipeline on the Ames Housing Dataset — from data cleaning and feature engineering to explainable AI and Streamlit deployment.

1,460
Samples

80
Features

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Introduction & Problem Statement

Why It Matters

Property prices depend on interacting factors — location, size, quality, age, and facilities. Traditional valuation relies on manual inspection and expert judgment, which is time-consuming and subjective. Machine learning offers a data-driven alternative that learns these complex relationships directly from historical sales data.

Target Variable

SalePrice

The final sale price of each residential property, in US dollars.

Project Objectives

1. Clean and preprocess housing data
2. Analyze patterns using EDA
3. Engineer meaningful features
4. Train multiple regression algorithms
5. Optimize via hyperparameter tuning
6. Evaluate and compare models
7. Explain predictions with SHAP
8. Deploy as an interactive web app

Dataset Description

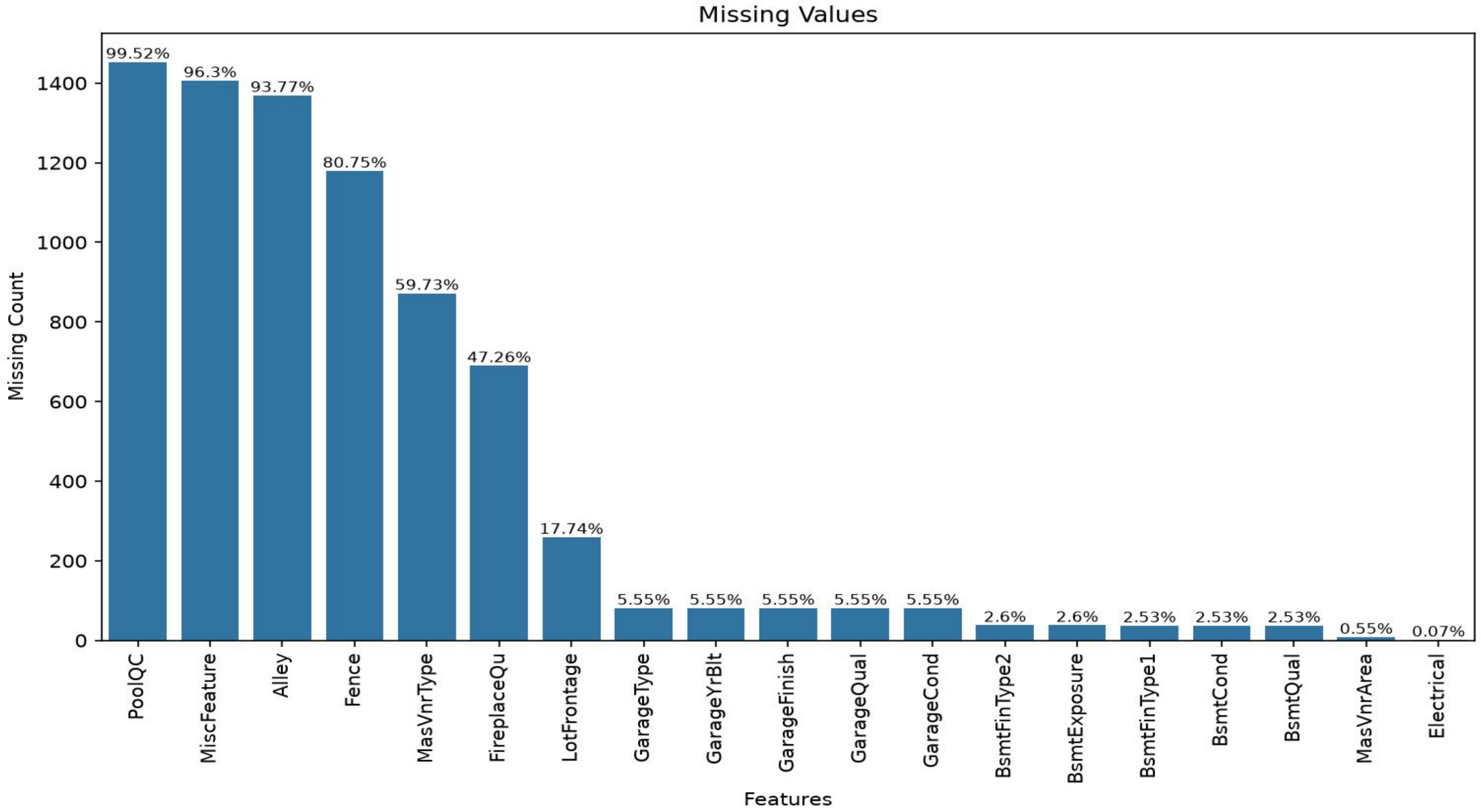
Ames Housing Dataset — a widely used regression benchmark

1,460	81	38	43
Training Samples	Original Features	Numerical Features	Categorical Features

What the Data Contains

- Comprehensive property information: structural characteristics, quality ratings, neighborhood data, garage and basement details.
- No duplicate records were found — every observation represents a unique residential property.
- Several features contain missing values, mostly tied to optional characteristics such as pools, fences, alleys, fireplaces, garages, and basements.

Data Cleaning — Missing Value Analysis



Key Findings

Alley, PoolQC, Fence, MiscFeature had more than 80% missing values and were removed — limited predictive value, insufficient data for imputation. This 4 features were removed

Handling Strategy

"None"	Categorical features where absence means the feature doesn't exist (BsmtQual, GarageType, FireplaceQu, etc.)
Median	LotFrontage, MasVnrArea, GarageYrBlt — robust to extreme values
Mode	Electrical — only a few missing values

Data Cleaning

1 Low-Information Features Removed

Street, Utilities, Condition2, RoofMatl — near-zero variance with almost all values in a single category.

2 Outlier Removal

Removed properties with GrLivArea > 4,000 sq ft and SalePrice < $300,000 — unusually large homes sold at low prices that could distort regression models.

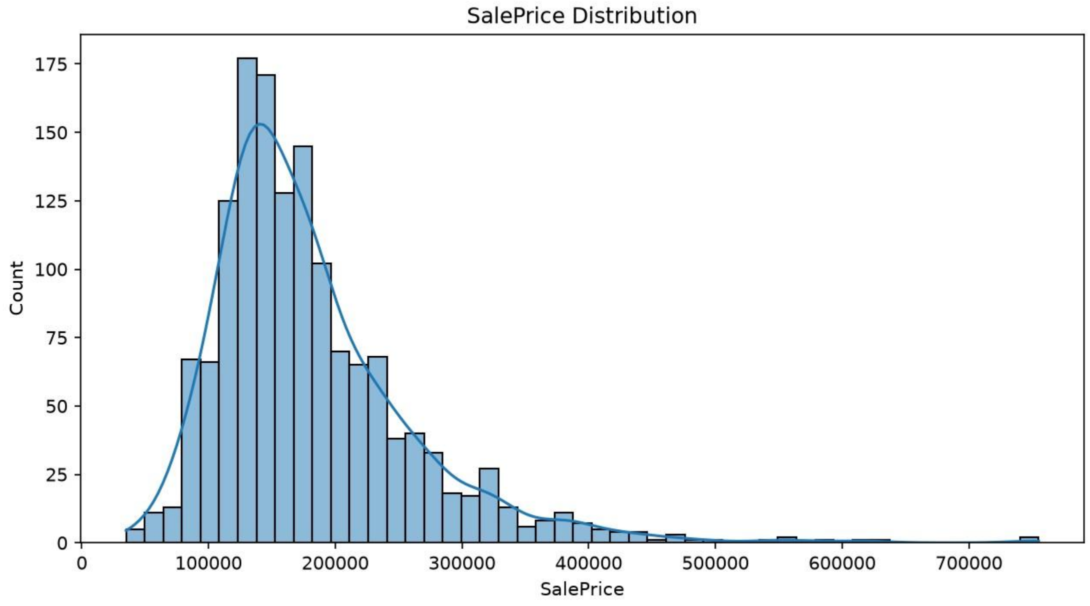
3 Feature Encoding

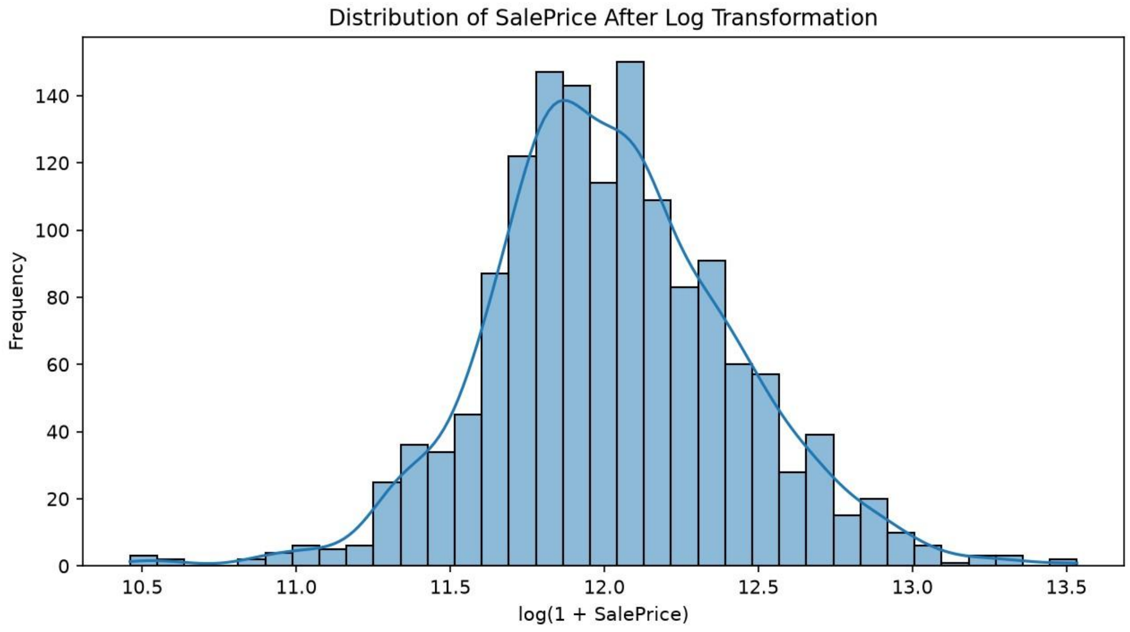
One-Hot Encoding for categorical variables.

4 Feature Transformation Pipeline

A Scikit-learn `ColumnTransformer` and `Pipeline` were used to perform median imputation and feature scaling for numerical features

Exploratory Data Analysis — SalePrice Distribution

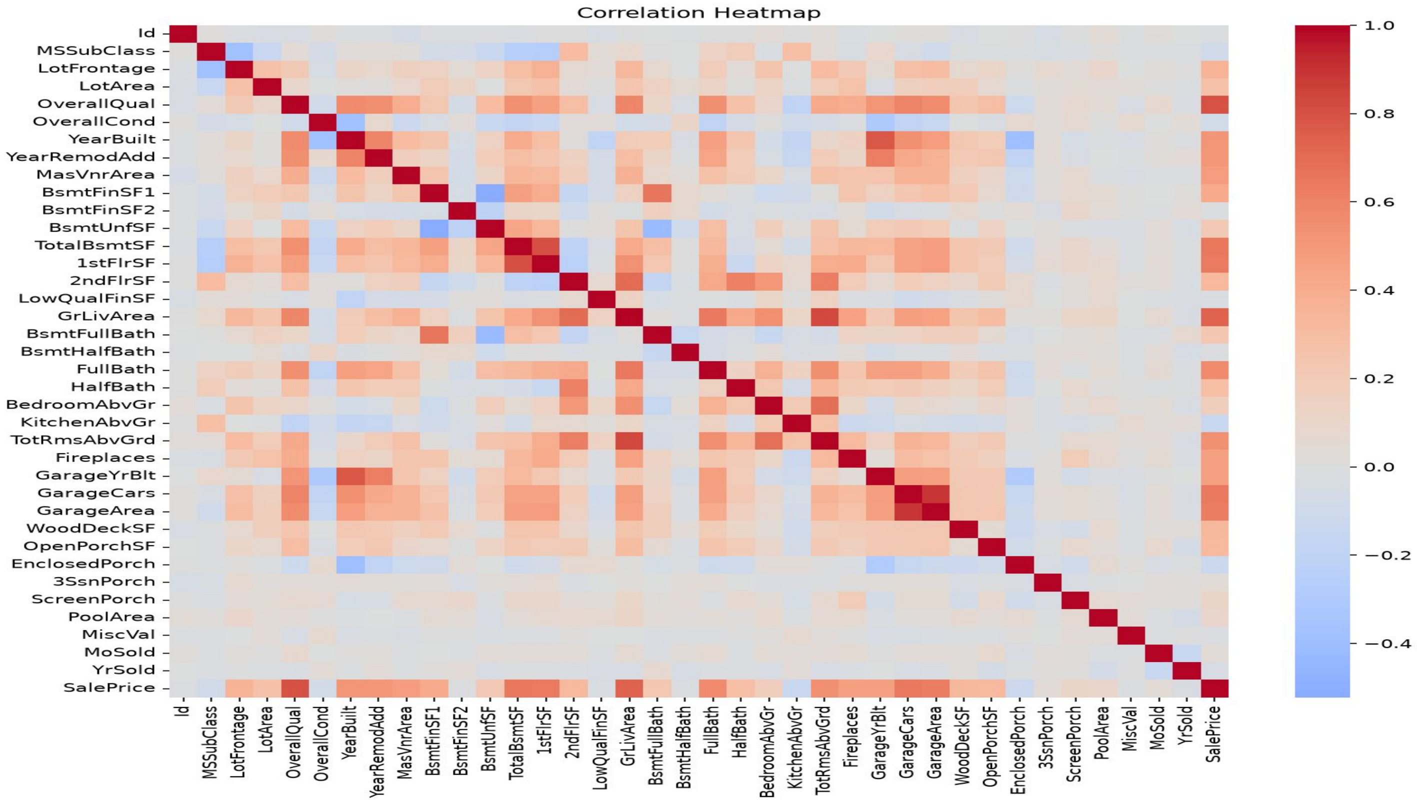




House prices are right-skewed

SalePrice shows a typical right-skewed distribution common to real-estate data. A log transformation, log(1 + SalePrice), was analyzed to normalize the distribution and better support model assumptions and visualization.

Exploratory Data Analysis — Correlation Analysis



Strongest Predictors

OverallQual

Overall material & finish quality

GrLivArea

Above-ground living area (sq ft)

TotalBsmtSF

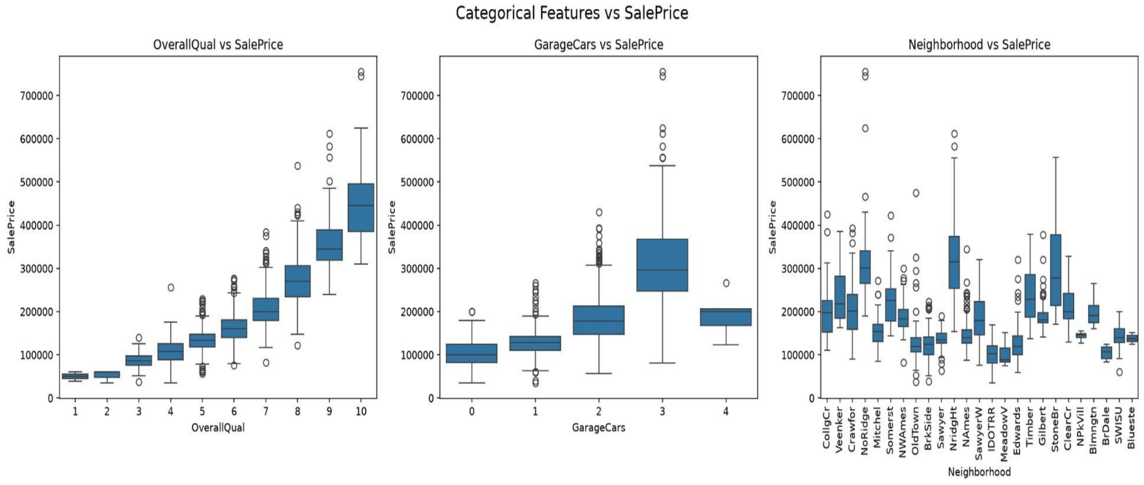
Total basement area

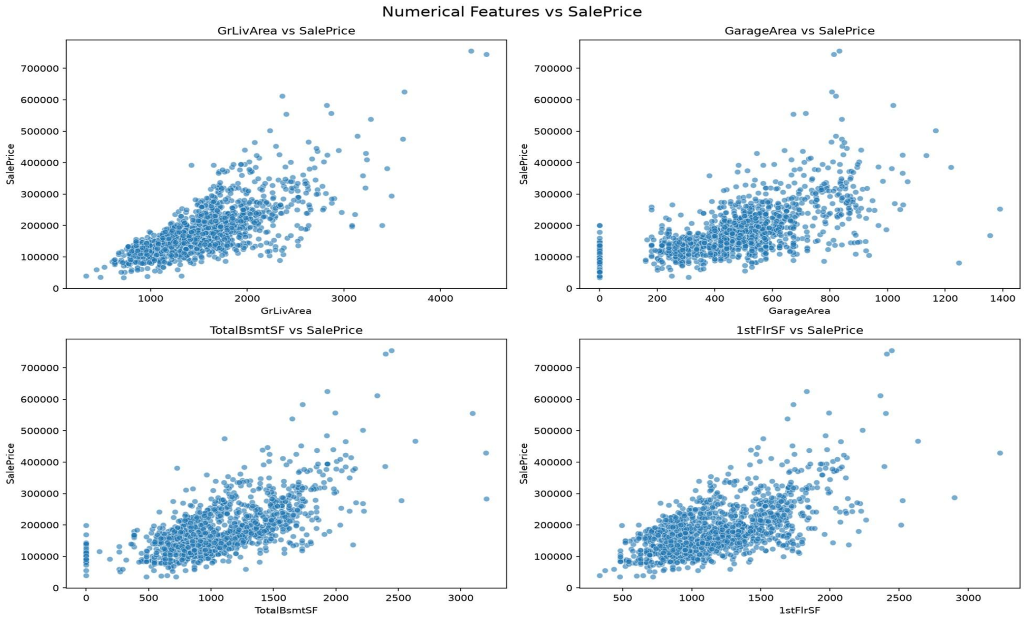
GarageCars

Garage capacity (# of cars)

These features were prioritized during feature engineering and model training.

EDA — Numerical & Categorical Relationships





Categorical Features vs SalePrice

Numerical Features vs SalePrice

Scatter plots of GrLivArea, GarageArea, TotalBsmtSF, and 1stFlrSF revealed clear positive relationships with SalePrice. Box plots of OverallQual, GarageCars, and Neighborhood showed that quality indicators and location strongly affect price distributions — properties with better materials, garages, and finished basements tended to sell higher.

Key EDA Findings

What the data told us

House quality is king

OverallQual was one of the strongest predictors of property value.

Size matters

Larger living areas generally corresponded to higher house prices.

Quality compounds

Better materials, garages, and finished basements meant higher selling prices.

Location & quality diverge

Neighborhood and quality attributes showed significant differences in price distributions.

Feature Engineering

Turning raw attributes into higher-signal predictors

Original Features	Engineered Feature
TotalBsmtSF + 1stFlrSF + 2ndFlrSF	TotalSF — total square feet
FullBath + 0.5×HalfBath + Bsmt baths	TotalBathrooms — weighted bathroom count
YrSold − YearBuilt	HouseAge — age at time of sale
YrSold − YearRemodAdd	YearsSinceRemodel — time since renovation
Porch & deck features combined	TotalPorchSF — total outdoor living space
OverallQual × GrLivArea	QualityLivingArea — quality-size interaction
GarageCars × GarageArea	GarageScore — garage capacity & size combined

QualityLivingArea and TotalSF emerged as the two most influential engineered features in the final model.

Regression Algorithms Evaluated

Linear, tree-based, ensemble, and gradient boosting approaches

Linear Regression

Baseline model

Ridge Regression

L2 regularization, α = 1.0

Lasso Regression

L1 regularization, α = 0.001

Decision Tree

Recursive feature splitting

Random Forest

200 trees, ensemble variance reduction

Extra Trees

Extra randomness for generalization

Gradient Boosting

Sequential error correction

AdaBoost

Focuses on difficult samples

SVR

HYperplane

XGBoost

400 estimators, tuned depth & subsampling

LightGBM

500 estimators, efficient tree boosting

CatBoost ⭐

Optimized for categorical & tabular data

Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation on CatBoost

Parameter	Search Values
Iterations	500(a chain of 500 trees)
Learning Rate	0.03, 0.05, 0.07
Tree Depth	3, 4
L2 Regularization	3, 5

Hyperparameter tuning improved performance by reducing prediction error and slightly increasing explained variance compared with the default CatBoost configuration.

Selection Criteria

The best configuration was chosen based on the highest cross-validation R^2 score across all parameter combinations.

Result

Tuned CatBoost outperformed the default configuration

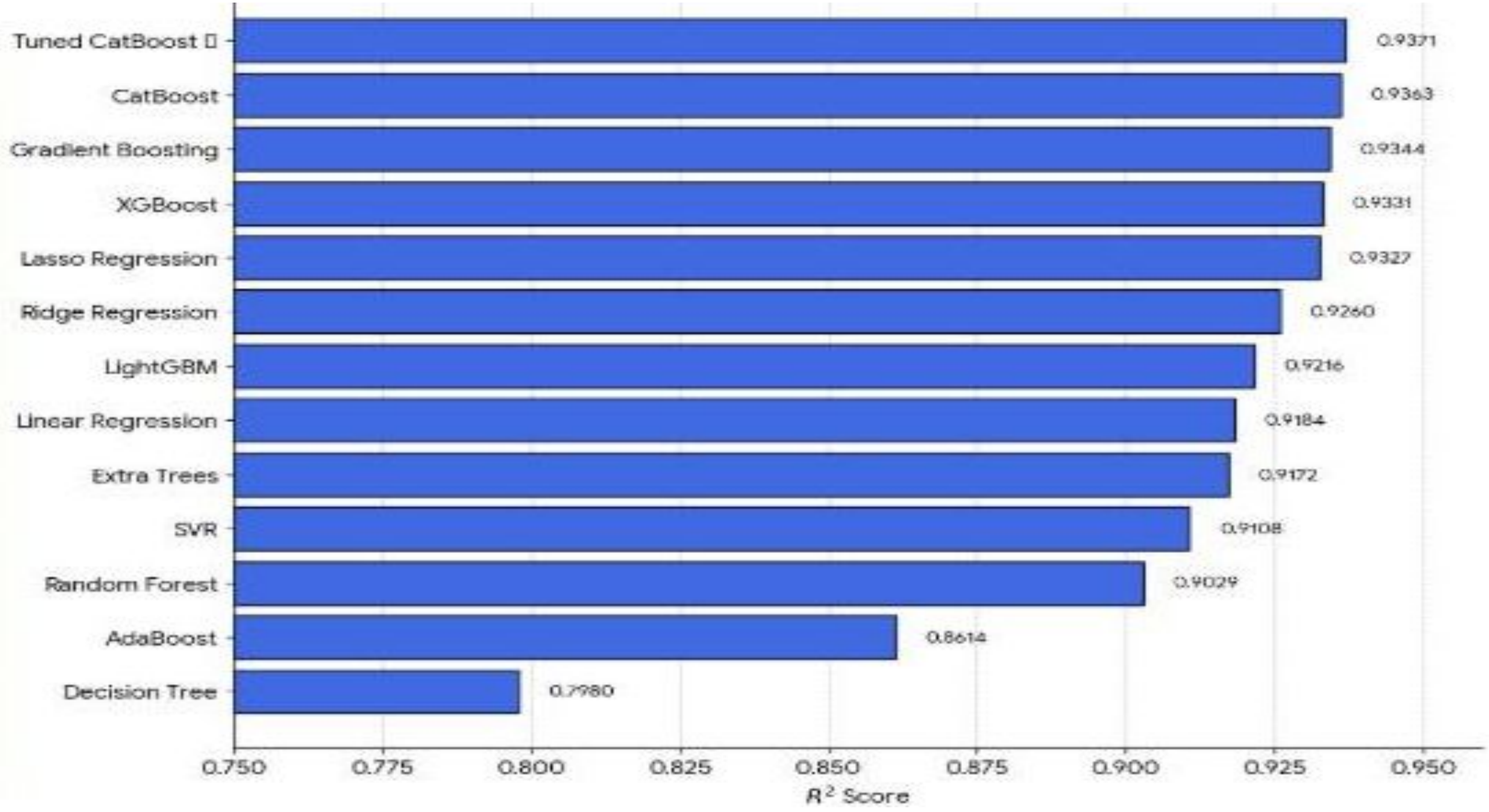
R^2 Score: **0.9363 ⟶ 0.9371**

RMSE: **18,765.32 ⟶ 18,636.19**

MAE: **13,721.36 ⟶ 13,680.99**

Model Comparison

R^2 Score across all 12 evaluated algorithms



Model Evaluation

Four standard regression metrics guided model comparison

MAE

Mean Absolute Error — average absolute difference between actual and predicted prices.

MSE

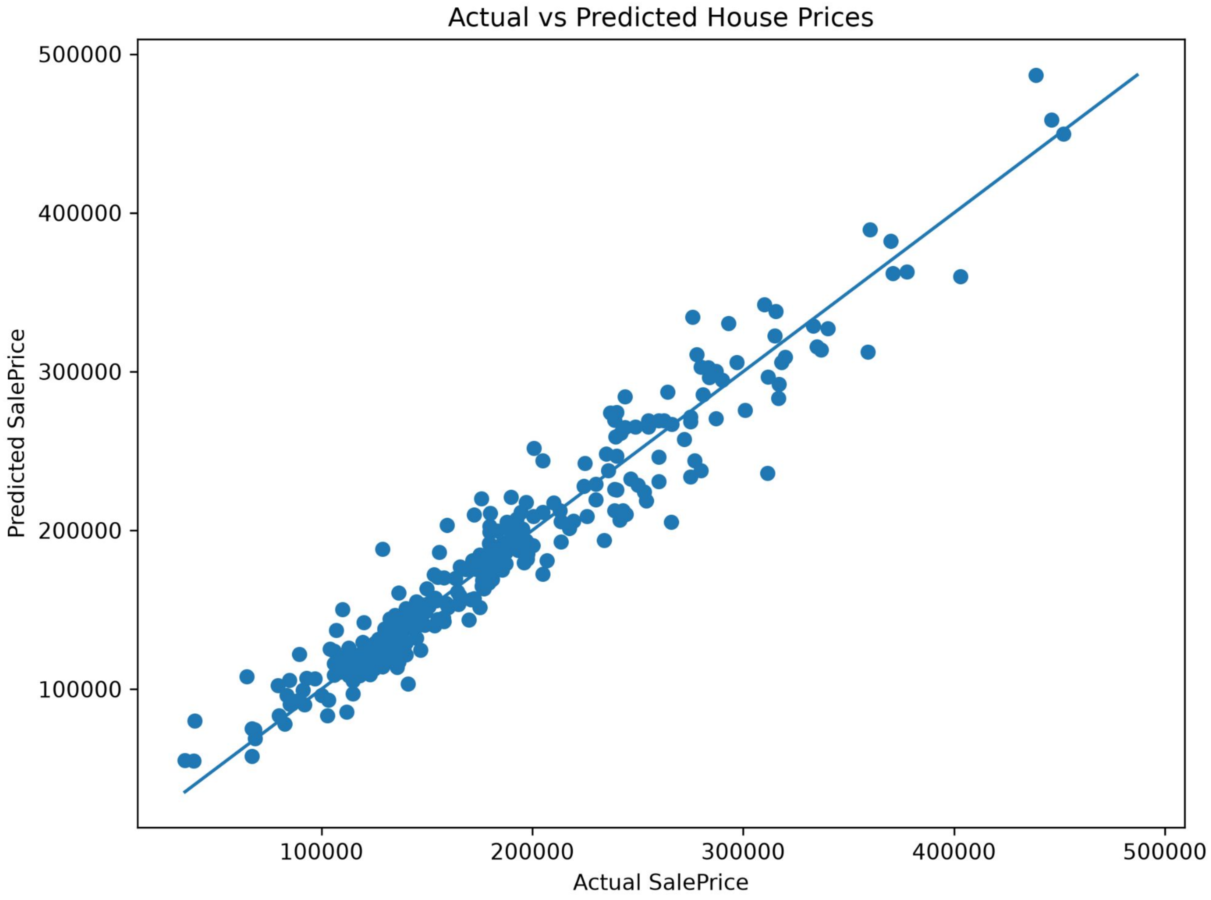
Mean Squared Error — squared differences; penalizes larger errors more heavily.

RMSE

Root Mean Squared Error — prediction error in the original price scale.

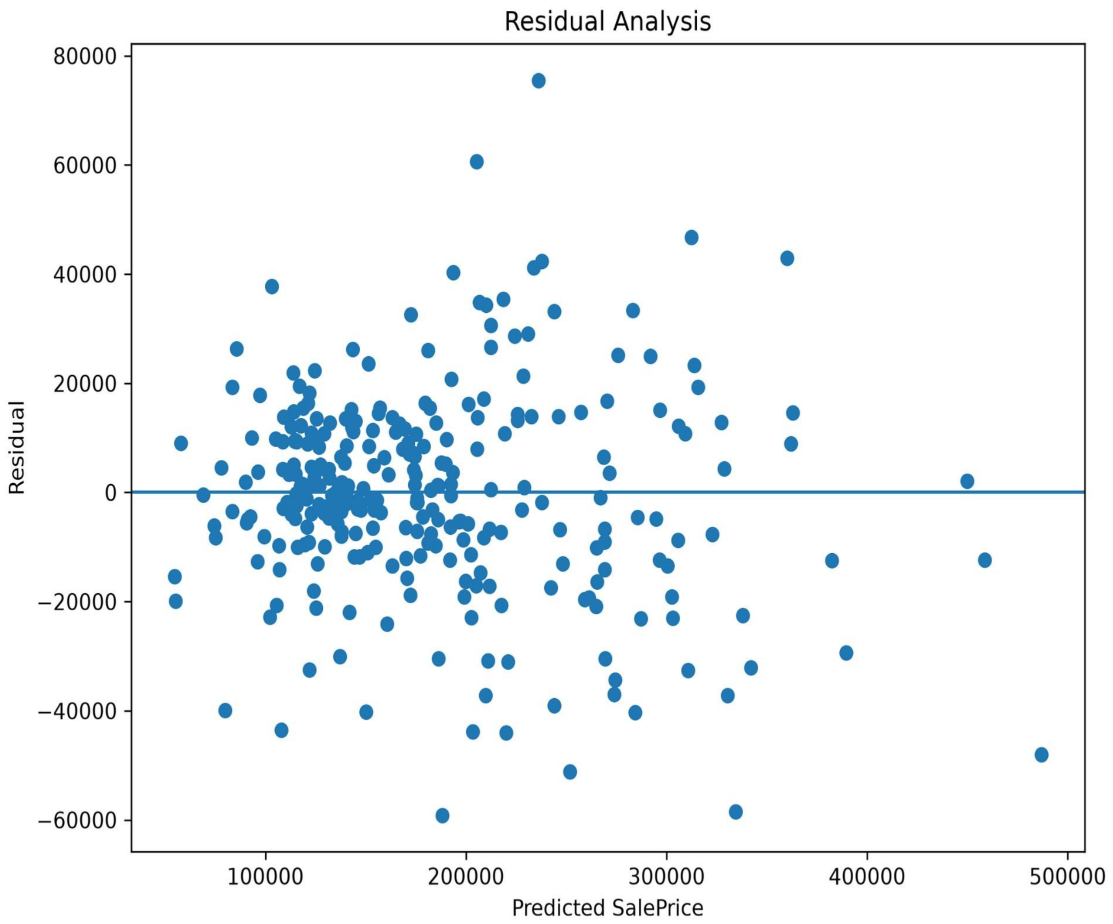
R^2

Proportion of variance explained; higher values indicate stronger predictive power.

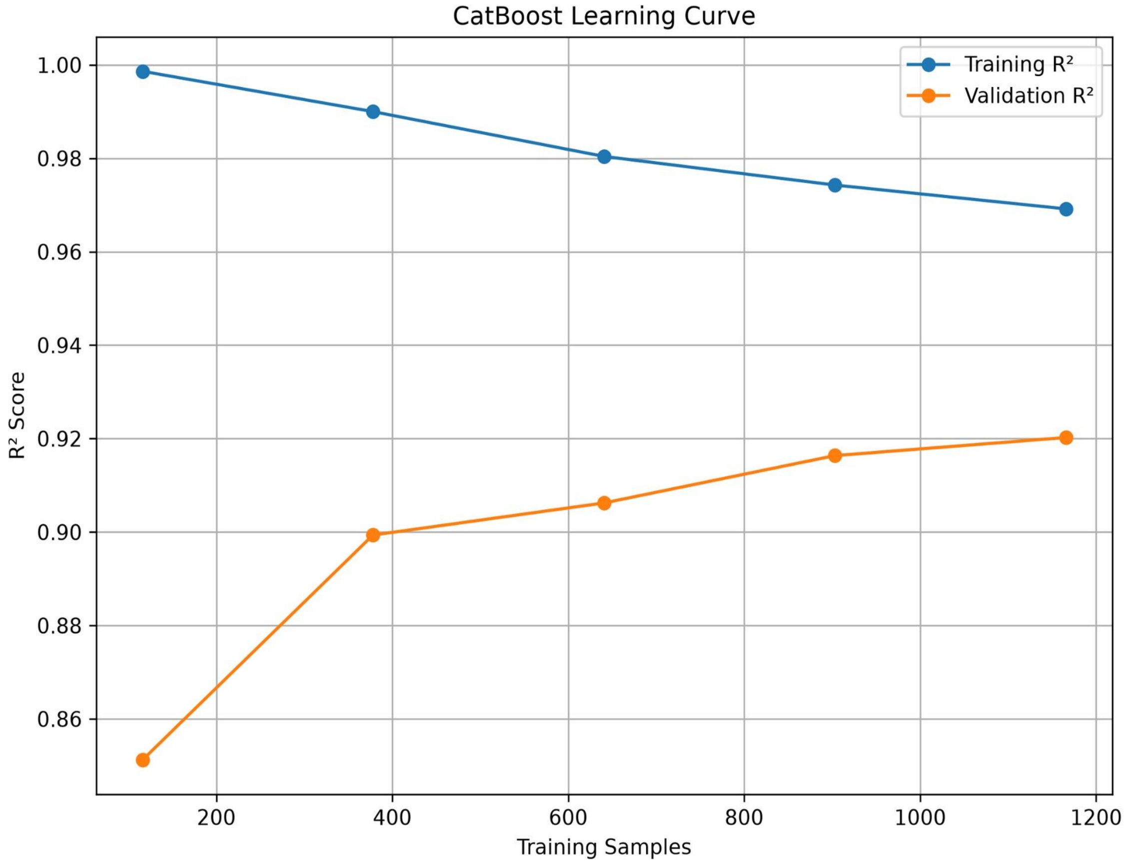


Actual vs Predicted House Prices — tight clustering along the diagonal confirms strong agreement.

Residual & Learning Curve Analysis



Residuals are randomly distributed around zero — no systematic prediction bias.



Training and validation R^2 converge as sample size grows — good generalization, minimal overfitting.

Best Model Selection

Tuned CatBoost Regressor — the final selected model

0.9371

R^2 Score

Highest explained variance

13,680.99

MAE (USD)

Lowest mean absolute error

18,636.19

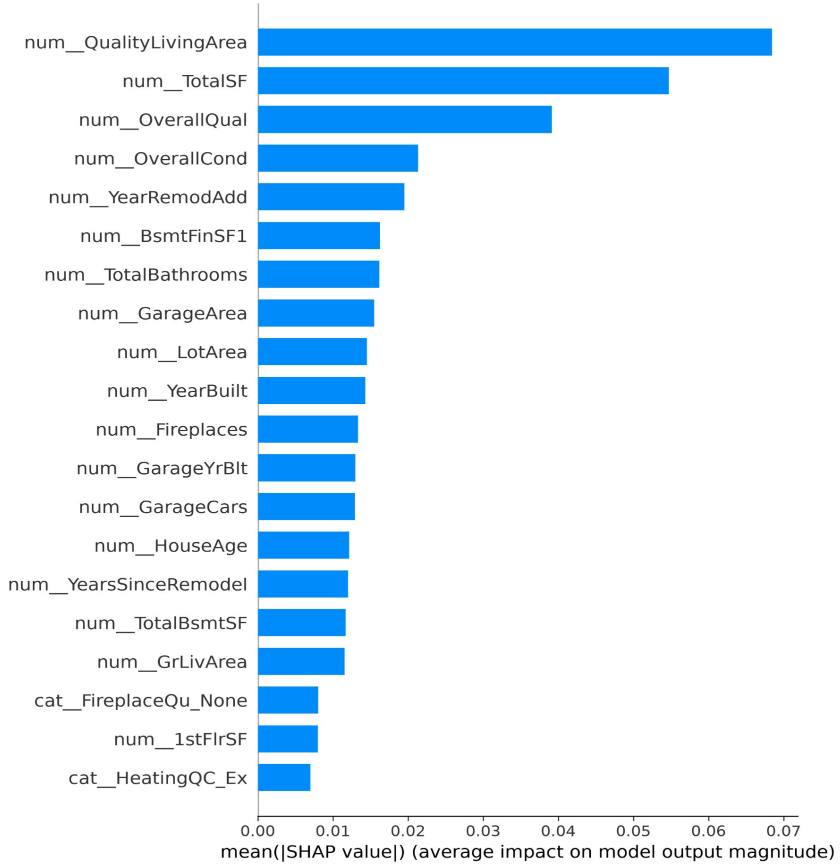
RMSE (USD)

Lowest root mean squared error

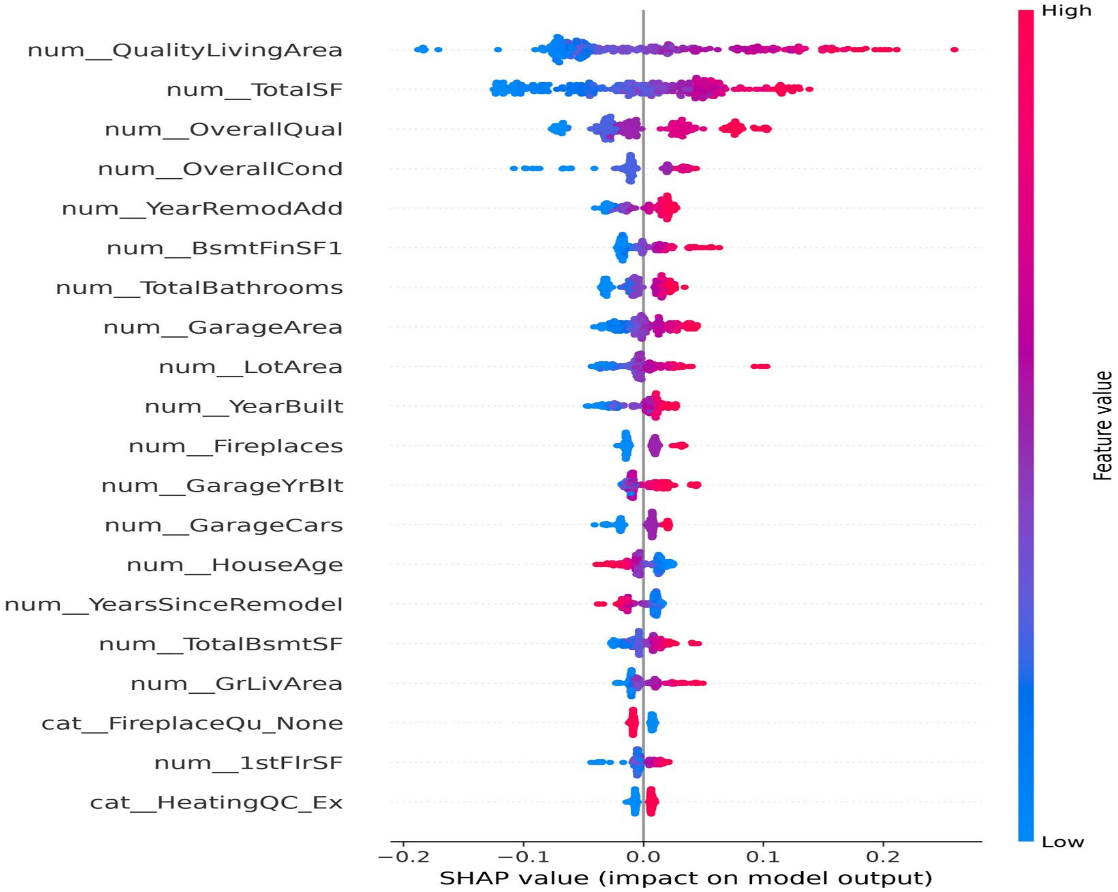
Hyperparameter tuning improved the CatBoost model over its default configuration — reducing prediction error and slightly increasing explained variance. The final tuned model was saved and integrated into the Streamlit prediction application.

SHAP Explainability Analysis

Shapley Additive exPlanations — global feature impact & direction



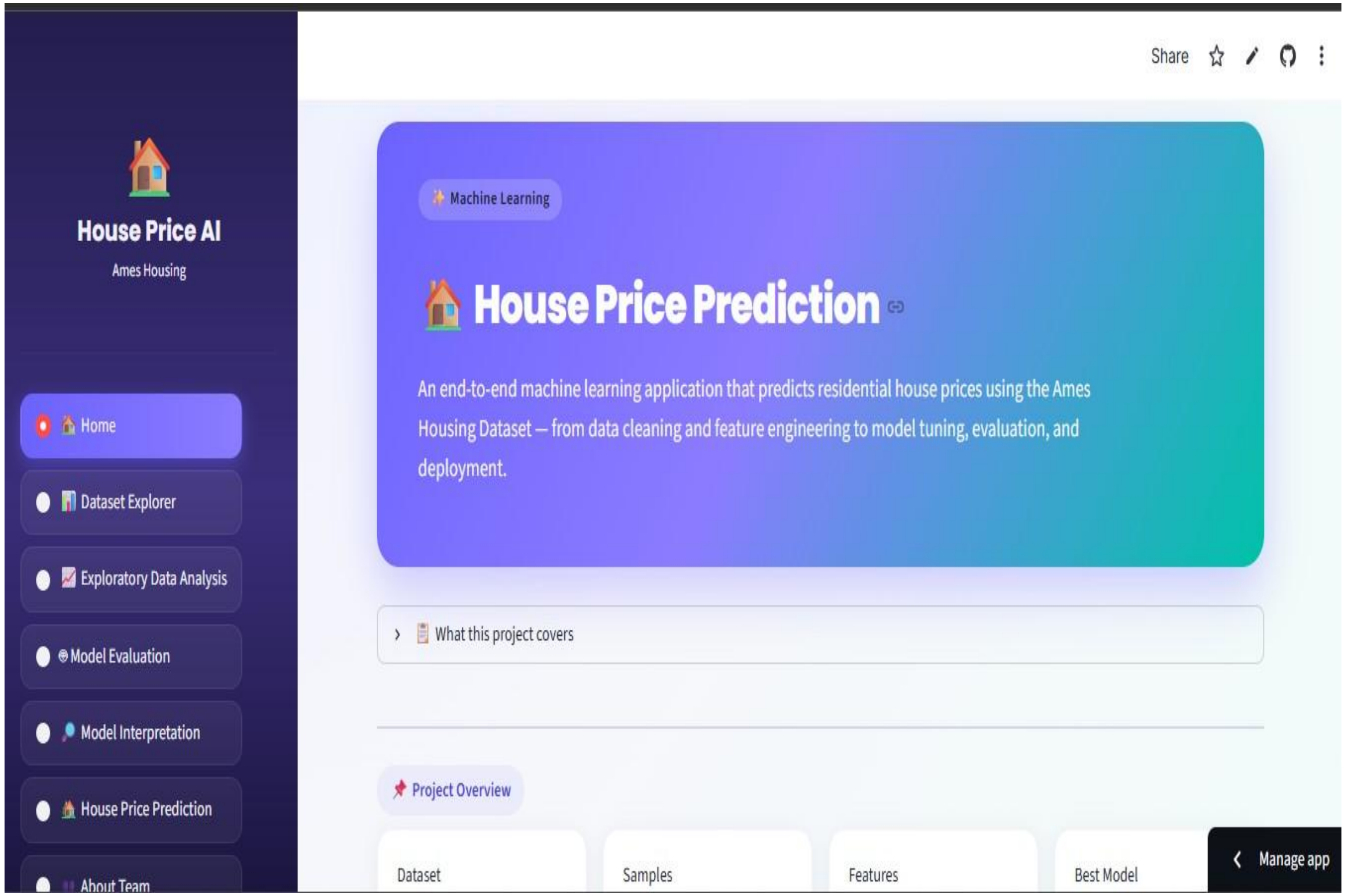
SHAP Feature Importance — average contribution



SHAP Summary — direction of influence (red = high value)

Deployment

Interactive Streamlit application, containerized with Docker



App Features

- ✓ Dataset Exploration
- ✓ Interactive EDA visualizations
- ✓ Model Evaluation Dashboard
- ✓ Feature Importance Analysis
- ✓ SHAP Explainability
- ✓ Interactive Price Prediction

Deployment Stack

Streamlit Cloud (hosting) · Docker (containerization) · Docker Hub (image registry)

Conclusion

This project successfully developed an end-to-end machine learning regression system for residential house price prediction using the Ames Housing Dataset — covering the complete workflow from data understanding to deployment.

1. **Data Understanding**
2. **Cleaning & Preprocessing**
3. **Exploratory Analysis**
4. **Feature Engineering**
5. **Model Development**
6. **Hyperparameter Tuning**
7. **Evaluation & Comparison**
8. **Explainability**
9. **Deployment**

The final system provides both accurate house price predictions and interpretable insights into model decision-making — a complete machine learning solution from data processing to deployment.

Future Improvements

1 **Advanced Hyperparameter Optimization**

Bayesian Optimization, Optuna, or automated search strategies for further gains.

2 **Cloud Deployment Expansion**

Deploy via AWS, Microsoft Azure, or Google Cloud for scalability and accessibility.

3 **Automated Machine Learning (AutoML)**

Automatically explore preprocessing, feature engineering, and model selection.

4 **Model Monitoring & Maintenance**

Track performance, detect data drift, and schedule model updates.

5 **Real-Time Market Integration**

Incorporate external housing market data to improve prediction accuracy.

6 **MLOps Implementation**

Automated training, testing, version control, model registry, continuous deployment.

Thank You

House Price Prediction — Ames Housing Dataset